Auditory N1 is modulated by alphabetic literacy skills in a meta-phonological task

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Abstract

Learning an alphabetic orthography enhances phonological awareness. However, it remains unclear how alphabetic literacy skills influence different processing stages involved in a meta-phonological task. Besides, few studies have investigated this 'processing stage' issue by comparing logographic readers with vs. without expertise in an alphabetic script. We recruited Cantonese-speaking Chinese readers with high and little proficiency in *Jyutping*, a Romanization system of Cantonese, and examined their event-related potentials elicited by speech and pure-tone stimuli during an onset discrimination task. Individuals with high *Jyutping* skills achieved higher accuracy and exhibited larger auditory N1 amplitude than those with low *Jyutping* skills in the phoneme (speech) discrimination task. No group effects were found in the pure-tone task, indicating that benefits of alphabetic literacy skills might be restricted to speech. We propose that the enhanced N1 amplitude reflects improved phonetic/phonological processing, which may be a consequence of enhanced selective attention to phonemic units in Chinese readers with high *Jyutping* skills. These findings have implications for understanding neuro-cognitive changes induced by literacy acquisition and the building blocks of phonological awareness.

Keywords: Phonological awareness; alphabetic orthography; auditory evoked potentials; Cantonese; *Jyutping*

1. Introduction

For languages in a literate culture, there is a bi-directional association between orthography and phonology (Blachman, 2000; Bus & van IJzendoorn, 1999; Ziegler & Ferrand, 1998). On the one hand, previous research reveals that phonological knowledge exerts substantial influence on the literacy development (Bus & van IJzendoorn, 1999; Wagner et al., 1987) and visual word recognition (Stone & Vanhoy, 1997). On the other hand, accumulating evidence indicates that literacy acquisition transforms speech processing in a significant way (Dehaene et al., 2010, 2015; Monzalvo & Dehaene-Lambertz, 2013; Morais et al., 1979; Ziegler & Ferrand, 1998).

One piece of evidence for the orthography-to-phonology coupling comes from studies examining the on-line influence of orthographic representations on phonological processing in spoken word recognition (Pattamadilok et al., 2009; Perre et al., 2009; Perre & Ziegler, 2008; Ventura et al., 2004; Ziegler et al., 2008; Ziegler & Ferrand, 1998). Most of these studies mainly manipulated one specific factor, i.e., the orthographic consistency of spoken words, and unveiled negative impact of orthographic inconsistency in spoken word recognition. In auditory lexical decision tasks where participants are required to determine the lexicality of an auditory input, consistent words, whose rimes can only be spelled in one way (e.g., /-ʌk/: -uck), typically engender shorter response latencies and fewer errors than inconsistent words, whose rimes have multiple spellings (e.g., /-i:p/: -eap, -eep). Event-related potential (ERP) research further showed that the consistency effect emerges as early as 350 ms after stimulus onset when the consistency of the first syllable in a word is manipulated. This suggests an early influence of orthographic representations on spoken word recognition before lexical access (Pattamadilok et al., 2009; Perre et al., 2009; Perre & Ziegler, 2008). Regarding the brain regions responsible for the consistency effect, ERP source analysis revealed a left temporo-parietal region subserving phonological processing (Perre et al., 2009).

Another critical piece of evidence in support of the orthography-to-phonology connection is the close relationship between meta-phonological ability, i.e., phonological awareness, and literacy acquisition. Taken literally, phonological awareness is an awareness of phonological units, e.g., phonemes, onset/rimes and syllables, in the speech stream (Blachman, 2000; Wagner et al., 1987). It typically refers to the ability to identify and manipulate phonological units in spoken language, as measured by various meta-phonological tasks, such as phoneme deletion and addition (de Gelder et al., 1993; Huang & Hanley, 1995, 1997; Morais et al., 1979; Read et al., 1986), non-word repetition (Reis & Castro-Caldas, 1997), sound unit judgement (de Gelder et al., 1993), sound matching (also called sound detection, Cheung et al., 2001; Ho and Bryant, 1997; Shu et al., 2008), and phoneme counting (Liberman et al., 1974).

The psycholinguistic grain size theory (Ziegler & Goswami, 2005) provides a powerful theoretical framework for understanding the relationship between phonological awareness and literacy acquisition across different languages. This theory was put forward to account for cross-linguistic differences in phonological development, reading development, skilled reading and developmental dyslexia. A hallmark feature of this theory is its emphasis on the use of different psycholinguistic grain sizes in phonology-orthography mapping, e.g., phonemes, onsets/rimes (onset/nucleus/coda), syllables, and whole words, for different language systems. More specifically for phonological development, the theory addresses influences of both spoken language characteristics, like phonological structure, (Cheung et al., 2001; O'Seaghdha et al., 2010; Roelofs, 2014) and literacy acquisition (Liberman et al., 1974; Morais et al., 1979; Reis & Castro-Caldas, 1997) on phonological awareness. Cross-linguistic differences in literacy-induced enhancement of phonological awareness are assumed to arise as a consequence of different psycholinguistic grain sizes employed in the world's writing

systems (Perfetti et al., 2010; Ziegler & Ferrand, 1998). In an alphabetic writing system like Dutch, the orthography represents speech sounds at a more fine-grained, i.e., phonemic, level, whereas, in a logographic system like Chinese, orthography is associated with phonology at a much coarser grain-size, i.e., the morpho-syllabic level. The phonology-orthography association is even more complicated for some mixed writing systems, like Japanese, where both syllabic or strictly speaking moraic 'kana' and logographic Chinese characters are utilized (Inagaki et al., 2000; Yoshihara et al., 2017). Due to differences in psycholinguistic grain sizes, the level of phonological awareness that is boosted by literacy acquisition depends on the nature of a specific script. Accordingly, the enhancement at the relatively fine-grained phonemic level should be more pronounced for individuals who are familiar with an alphabetic system. There is mounting behavioral evidence for this argument. It has been reported that children with less alphabetic literacy expertise (e.g., pre-reading children) have difficulties with explicit phoneme manipulation, compared with more advanced readers (e.g., Liberman et al., 1974). For adults with similar demographic characteristics, the illiterate showed lower performance in tasks tapping explicit phoneme analysis than literate alphabetic readers (Morais et al., 1979; Reis & Castro-Caldas, 1997). Cross-linguistic comparisons conducted by Ho and Bryant (1997) indicated that Chinese children in Hong Kong developed consonant awareness later than their English counterparts at primary school levels. The authors attributed this late development of consonant awareness to differences between logographic Chinese and alphabetic English writing systems. Other studies compared the phonological awareness skills in logographic Chinese readers with and without knowledge of alphabetic scripts, including a native *Pinyin* script for Mandarin Chinese (Read et al., 1986) or a second language Dutch script (de Gelder et al., 1993). These studies have consistently reported that Chinese readers with alphabetic knowledge are more capable of manipulating phonemes explicitly than those literate only in logographic Chinese. Additionally, it was also reported that *Zhu-yin-fu-hao*, an alphabetic script for Mandarin Chinese in Taiwan, boosted phonological awareness for logographic Chinese readers (Huang & Hanley, 1995, 1997).

Neuroimaging studies have provided further insight into the underlying cognitive processes and their corresponding brain networks involved in meta-phonological tasks, and how they are altered by alphabetic literacy acquisition (Brennan et al., 2013; Burton et al., 2000; Castro-Caldas et al., 1998; Monzalvo & Dehaene-Lambertz, 2013; Petersson et al., 2000; Zatorre et al., 1996). Despite varying task demands across these studies, the left-dominant peri-sylvian language network, comprised of frontal and temporo-parietal regions, has been found to be implicated in meta-phonological tasks and literacy-induced changes in speech processing. Moreover, an additional left occipito-temporal region, i.e., the posterior left fusiform gyrus or visual word form area (VWFA), subserving orthographic representations and processing, is added to the language network following literacy acquisition and may become activated in a top-down manner during speech comprehension (Burton et al., 2000; Dehaene et al., 2010, 2015). Finally, other networks, such as the attentional network involving anterior cingulate cortex and the central executive network involving prefrontal cortex, are also involved when meta-phonological analyses are required, and vary in functional activations according to alphabetic literacy skills (Burton et al., 2000; Petersson et al., 2000).

Burton et al. (2000) conducted an fMRI study to examine the contribution of the frontal cortices of the peri-sylvian language network during speech-discrimination tasks that required different levels of phonemic segmentation efforts. They created two lists of speech stimuli—'global' and 'segmental'—for an onset discrimination task, where participants were asked to make 'same/different' judgment on the initial consonants of English spoken word pairs. In the global list, as the rimes were the same in each word pair (e.g., dip-tip, dip-dip), participants were expected to adopt a 'global strategy', i.e., to make discriminations without segmenting

out the initial phonemes. Conversely, in the segmental list, the rimes of the two words differed (e.g., dip-tomb, dip-doom). Thus, to perform the task correctly, participants might have to separate initial phonemes from the rest of the words. The 'segmental strategy' could also be considered as involving more temporal selective attention mechanisms (Correa et al., 2005; Lange et al., 2003). For stimuli in this list, the participants had to direct more their temporal attention to the initial phoneme and ignore or inhibit interference from the subsequent rime as the spoken word unfolds in time. Besides the phoneme discrimination task, a pure tone discrimination task was employed as a control task by Burton and colleagues. The imaging results indicated that the superior temporal region, hypothesized to be responsible for translating acoustic signals into phonological categories, was activated for both lists. However, only the stimuli in the segmental list exhibited consistent activations in frontal cortices, which presumably reflected increased demands of phonemic segmentation, articulatory recoding or working memory in the segmental list. In addition, the authors found stronger activation in the inferior temporal lobe (Brodmann 37) in the segmental list than in the global list, which they interpreted as being caused by the co-activation of orthographic (visual) representations while segmenting the speech stimuli. Finally, stronger activation in the anterior cingulate cortex was reported in the segmental list, possibly reflecting increased attentional demands or conflict monitoring in this list. In summary, these results have provided crucial insight into the cognitive processes and their corresponding brain networks in meta-phonological tasks, but they cannot reveal how these processes unfold in time in online speech processing.

There are also some ERP studies investigating the time course of orthographic effects in meta-phonological tasks (e.g., Lafontaine et al., 2012; Pattamadilok et al., 2011). For example, Lafontaine et al. (2012) manipulated the congruency (- or +) between the phonological (P) and orthographic information (O) of the initial phonemes of pairs of French spoken words, resulting in four conditions—P+O+, e.g., /ʒilɛ/-/ʒənu/, gilet-genou, 'vest-knee'; P+O-, e.g., /ʒɑ̃bɔ̃/-/ʒənu/, jambon-genou, 'ham-knee'; P-O-, e.g., /kɔmɑ̃/-/ʒənu/, comment-genou, 'how-knee'; P-O+, e.g., /gato/-/ʒənu/, gateau-genou, 'cake-knee'. These materials were presented in an onset discrimination task where participants were asked to make 'same/different' judgement on the onset phonemes. Their results revealed the time course of activation of phonological and orthographic information during a meta-phonological task: phonological effects occurred at 250~300 ms after stimulus onset and gradually disappeared later on (300~500ms) whereas the orthographic effects first emerged and interacted with phonology at 300~500 ms and became predominant at a later post-lexical processing stage (500~700 ms). However, these ERP studies focuses on the congruency between phonology and orthography and did not address the question how the development of phonological awareness induced by alphabetic literacy skills influences speech processing in a meta-phonological task.

The main purposes of the current study were to investigate the impact of learning a native alphabetic script in adulthood on phonemic awareness skills in logographic Chinese readers, and, more importantly, to identify the speech processing stage(s) involved in a meta-phonological task that is/are sensitive to alphabetic literacy skills. Precisely, the present study extended the existing literature on phonological awareness by examining the impact of learning *Jyutping*, an alphabetic system that represents spoken Cantonese in Hong Kong (Linguistic Society of Hong Kong, 1997), on phonemic awareness in Cantonese-speaking logographic Chinese readers. Unlike *Pinyin* or *Zhu-yin-fu-hao* for Mandarin Chinese, which is typically taught to pre-reading children to assist the learning of Chinese characters, no Chinese Romanization system is included in the school curriculum in Hong Kong. Most *Jyutping* users have acquired this alphabetic script through self-instruction in adulthood for inputting Chinese characters on a computer or for other pedagogical purposes. This provides an opportunity to examine whether and to what extent the relatively late acquisition of an

alphabetic script will influence the neuro-cognitive processes that underpin phonological awareness skills in one's native language, despite the fact that most Cantonese speakers in Hong Kong already have some acquaintance with the alphabetic English script.

To this end, we recruited a group of Cantonese-speaking logographic readers with high *Jyutping* competency and a control group with little *Jyutping* knowledge. Their phonemic awareness skills were examined through a phoneme discrimination task in which participants were required to judge whether two Cantonese pseudowords shared the same initial phoneme. In order to examine whether this segmentation ability is restricted to language materials, a pure tone discrimination task, where participants had to judge whether two tone-triplets shared the same initial tone, was used as a control task, similar to the design of Burton et al. (2000). ERP responses were examined during both tasks. As in Burton et al.'s study (2000), the experimental materials were separated into 'global' and 'segmental' lists, which presumably required different segmentation efforts and attentional demands. This manipulation would allow us to examine not only how the neuro-cognitive processes differ between the two lists but also how list differences may interact with alphabetic literacy skills.

Based on existing behavioral literature (de Gelder et al., 1993; Read et al., 1986), we predicted that *Jyutping* skills would enhance behavioral responses in the phoneme discrimination task, leading to higher response accuracy and shorter reaction time. For ERP responses, we expected that the auditory stimuli would elicit early auditory evoked potentials like N1-P2 (Lafontaine et al., 2012). As N1-P2 or its magnetic counterpart has been reported to code phonetic features and reflect early stages of phonetic/phonological processing (Bonte & Blomert, 2004; Obleser et al., 2003), we predicted that it might serve as a potential ERP marker of phonological awareness or alphabetic literacy skills. More specifically, we predicted that the *Jyutping* group would show stronger N1-P2 response than the control group, as skilled alphabetic readers have been found to show enhanced phonetic/phonological processing (Brennan et al., 2013; Dehaene et al., 2010, 2015). Furthermore, some later components such as the N2 and P3, which are associated with task-relevant higher-level cognitive processes, might also be elicited in the current paradigm (Folstein & Van Petten, 2008; Lafontaine et al., 2012; Polich, 2007; Zhang, 2017), and vary according to *Jyutping* skills. For list effects, based on Burton et al., (2000), there would be a difference in behavioral and ERP responses for 'global' vs. 'segmental' lists. Moreover, we hypothesized that there might be an interaction between *Jyutping* skills and list, i.e., the group difference in behavioral and ERP responses might be larger for the segmental list, which has higher segmentation and attentional demands. However, no influence of *Jyutping* knowledge was expected on either behavioral or neural responses in the pure tone discrimination task, which we included as a control task in the current study.

2. Methods

2.1. Participants

A *Jyutping* screening test was administered to all the participants who signed up for the experiment to assess their *Jyutping* proficiency. The test included 20 disyllabic Chinese words, comprising 40 Chinese characters, and provided complete coverage of Cantonese phonemes. The participants were required to transcribe the Chinese words into *Jyutping* within 2 minutes. Participants earned one point for correctly transcribing one character and thus could obtain a maximum score of 40. Twenty-seven participants with high *Jyutping* competency (henceforth, *Jyutping* group) and 25 participants with little *Jyutping* knowledge (henceforth, control group) were selected for the electroencephalographic (EEG) experiment. The 27 participants in the *Jyutping* group achieved a score of more than 20 points, while the scores of the remaining 25 participants in the control group were below 20 points. The *Jyutping* group and the control group were matched for English proficiency, musical

experiences and age. Their English proficiency was evaluated based on their most recent scores obtained in standardized English tests (IELTS, HKDSE-ENG or HKALE-ENG)[1]. Relevant characteristics of the two groups are summarized in Table 1. Most of the participants are university students in Hong Kong. All participants were right-handed and reported no speech, hearing, reading or other neurological disorders.

2.2. Stimuli and experimental design

2.2.1 Phoneme discrimination task

The phoneme discrimination task included 96 Cantonese pseudoword pairs, which were constructed from six sets of Cantonese pseudowords (see Supplemental Materials Table S1). Pesudowords were used to maximize processes as required by meta-phonological tasks and minimize lexical activation. Each set contained four monosyllabic pseudowords, which were phonotactically plausible in Cantonese, but there were accidental lexical gaps for the chosen syllables with Tone 1 (high level tone), Tone 2 (high rising tone) and Tone 4 (low falling tone). Only Tone 1, Tone 2 and Tone 4 were included in the stimuli, as these three lexical tones are considered as peripheral tones in Cantonese tonal space and consequently easier to identify (Peng et al., 2012). Within each set, four syllables (e.g., paa, paai, faa, faai) were formed by combining two initials (e.g., p, f) with two rimes (e.g., aa, aai), with the tone kept identical. For each of the six sets, 16 pseudoword pairs were constructed by combining each pseudoword with itself and then with the other three pseudowords. This step finally gave rise to a total of 96 pairs of pseudowords for all six sets.

Following the design of Burton et al. (2000), the resulting 96 pseudoword pairs were further divided into two lists, which represented two experimental conditions–'global' and 'segmental'. In the global list, two pseudowords in a pair (P_1 and P_2) had identical rimes, whereas the initial could be identical or different (e.g., paa^2-paa^2, $paai^2$-$paai^2$, faa^2-faa^2, $faai^2$-$faai^2$; paa^2-faa^2, faa^2-paa^2, $paai^2$-$faai^2$ $faai^2$-$paai^2$)[2]. In the segmental list, they had different rimes, whereas again, the initial could be identical or different (e.g., paa^2-$paai^2$, $paai^2$-paa^2, faa^2-$faai^2$, $faai^2$-faa^2; paa^2- $faai^2$, $faai^2$- paa^2, faa^2- $paai^2$, $paai^2$- faa^2). All speech materials were recorded in a sound-treated room by a native Cantonese-speaking female speaker. The duration of each pseudoword was normalized to 560 ms. Within each pseudoword pair, the inter-stimulus interval (ISI; offset to onset) was 440 ms.

2.2.2. Pure tone discrimination task

The pure tone discrimination task included 96 tone-triplet pairs, which were constructed from six sets of tone triplets (see Supplemental Materials Table S2). The tone triplets, consisting of three concatenated pure tones, were intended to mimic the temporal initial-rime structure of the speech stimuli. The 1st tone in the tone sequence was analogous to the initial consonant in the phoneme discrimination task while the 2nd and 3rd tones grouped together, constituting a tone-duplet, corresponded to the rime. The frequencies of the pure tones had little similarity to lexical tones or musical notes.

First, six pure tones were created for each set, e.g., 200_{Hz}, 1600_{Hz}, 800_{Hz}, 1000_{Hz}, 1400_{Hz} and 1800_{Hz} for set 1. Two tones were designated as the 1st tone in the tone-triplet, e.g., 200 Hz and 1000 Hz, while the other four tones were used to form two tone-duplets (2nd + 3rd tones), e.g., $1600_{Hz}800_{Hz}$ and $1400_{Hz}1800_{Hz}$. Then, for each set, four tone-triplets were constructed by combining two 1st tones and two tone-duplets (e.g., 200_{Hz}-$1600_{Hz}800_{Hz}$,

[1] IELTS, HKDSE and HKALE stand for *the International English Language Testing System*, *Hong Kong Diploma of Secondary Education Examination-English Language*, and *Hong Kong Advanced Level Examination-English Language* respectively. To make the standardized English tests scores comparable across participants, the HKDSE-ENG and HKALE-ENG scores were converted into equivalent IELTS band scores based on results of a benchmarking study comparing IELTS and HKDSE-ENG/HLALE-ENG by Hong Kong examinations and assessment authority (HKEAA).

[2] The Cantonese pseudowords are transcribed according to the *Jyutping* system. The superscript stands for Cantonese tones.

200_{Hz}-$1400_{Hz}1800_{Hz}$, 1000_{Hz}-$1600_{Hz}800_{Hz}$ and 1000_{Hz}-$1400_{Hz}1800_{Hz}$). Finally, for each of the six sets, 16 pairs of tone-triplets were generated by combining each tone triplet with itself and the other three tone-triplets. This step finally resulted in a total of 96 tone-triplet pairs.

As in the phoneme discrimination task, the 96 tone-triplet pairs were further classified into two lists. In the global list, two tone-triplets in a pair (T_1 and T_2) shared the same tone-duplets (i.e., 2nd + 3rd tones), whereas the initial tone could be identical or different, e.g., 200_{Hz}-$1600_{Hz}800_{Hz}$–200_{Hz}-$1600_{Hz}800_{Hz}$, 200_{Hz}-$1600_{Hz}800_{Hz}$–1000_{Hz}-$1600_{Hz}800_{Hz}$. In the segmental list, they ended with different tone-duplets, whereas again the initial could be identical or different, e.g., 200_{Hz}-$1600_{Hz}800_{Hz}$–200_{Hz}-$1400_{Hz}800_{Hz}$, 200_{Hz}-$1600_{Hz}800_{Hz}$–1000_{Hz}-$1400_{Hz}800_{Hz}$. The acoustic duration of the 1st tone was 280 ms while the 2nd tone and 3rd tone were 140 ms long each. This led to a total duration of 560 ms per tone-triplet. Within each tone-triplet pair, the inter-stimulus interval (ISI; offset to onset) was 440 ms.

2.3. Experimental Procedure

Participants were seated in a sound-treated and electrically-shielded booth. All instructions were given in Chinese to avoid or reduce possible influence of English knowledge on task performance and ERP responses. The whole experiment was divided into four blocks, representing four experimental conditions—speech/global, speech/segmental, pure tone/global, and pure tone/segmental. Each block lasted approximately 15 minutes and was preceded by 20 practice trials, in which feedback (reaction time and response accuracy) was provided. The auditory stimuli used in the 20 practice trials were created from two additional sets of pesudowords and tone-triplets, which were not presented in the experimental trials. All stimuli were repeated twice, giving rise to a total of 192 trials for each task. The order of speech vs. pure tone tasks and global vs. segmental lists within each task were counterbalanced across participants. Stimulus presentation and behavioral data collection were controlled by E-prime (Schneider et al., 2002).

Fig. 1 illustrates the experimental procedure of both the phoneme and pure tone judgment tasks. A similar onset discrimination paradigm was applied to both phoneme discrimination and pure tone discrimination tasks. A fixation cross, indicating the beginning of a trial, was presented at the center of the screen and remained until the end of a trial. 2000 ms after the onset of the fixation cross, a pair of stimuli was presented binaurally through insert headphones. Within each block, the stimuli were presented in a pseudorandom order. The participants were required to determine whether the two stimuli of the pair shared the same initial segment by pressing the left or right buttons of the response box with their left or right index finger. A response could be made during or after the delivery of the second stimulus of the pair without a time limit. The inter-trial interval was jittered between 1600 and 1800 ms. The assignment of hands (left and right) to same vs. different responses was counterbalanced across participants.

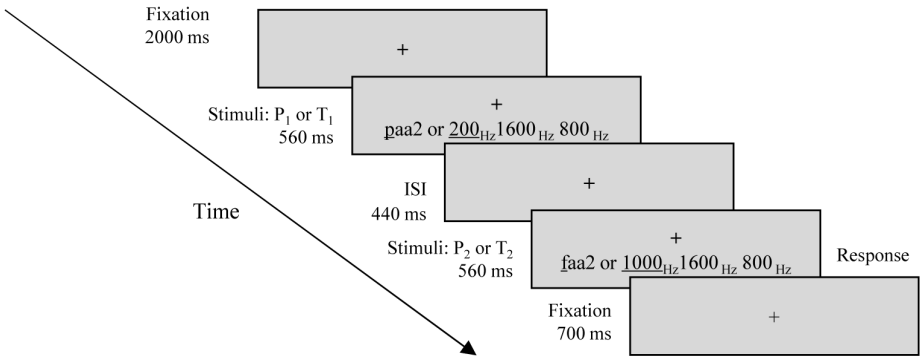

Fig. 1. The schematic flow diagram of one experimental trial. The underlined portions denote the initial consonants and the 1st pure tones, to which participants were required to make 'same' or 'different' responses.

2.4. EEG data acquisition and pre-processing

EEG signals were continuously recorded and digitized at a sampling rate of 1000 Hz using a 32-channel ActiveTwo EEG system (BioSemi B. V., Amsterdam, The Netherlands). Two electrodes (CMS/DRL) were used as an on-line reference. Vertical electrooculography (EOG) was recorded by two additional electrodes attached above and below the left eye to monitor vertical eye movement and blinks, and horizontal EOG was recorded by another two electrodes placed lateral to the external canthi of both eyes to monitor horizontal eye movements. Two additional electrodes attached over the right and left mastoids were used as the offline reference. Electrode impedance was kept below 5 kΩ.

To obtain the ERPs time-locked to the onset of the second pseudoword (P_2) and the second tone-triplet (T_2), the raw EEG data were preprocessed using BESA (Besa GmbH, Germany). We first applied a band-pass filter (1Hz - 30Hz) to the EEG data, and segmented the data into 900 ms epochs with a 100 ms pre-stimulus baseline and an 800 ms post-stimulus interval. Epochs with amplitudes exceeding ±75 μV at any electrode site were automatically rejected. On average, 84.9% and 72.6% of the trials were accepted for the *Jyutping* group and control group respectively. The EEG data from 5 participants (*Jyutping*: 2; control: 3) were excluded from further analysis due to excessive artifacts in the EEG recordings (acceptance rate < 50% for any experimental condition). The accepted epochs were averaged for each participant and each condition. The ERP data were recalculated against the average-mastoid reference.

2.5. Behavioral and ERP data analysis

The behavioral data from 25 *Jyutping* and 22 controls without excessive EEG artefacts were included in the analysis. Their response accuracy (ACC) and reaction time (RT) were analyzed using mixed-effects modelling. To test our prediction that *Jyutping* skills chiefly affect speech processing but exert little influence on auditory processing of non-linguistic materials, we fit two generalized linear mixed-effects models with a logit linked function to analyze ACC for the phoneme discrimination task and pure tone discrimination task respectively, and two linear mixed-effects models to analyze RT for the two tasks respectively. The four models included two deviation-coded (-0.5, 0.5) categorical variables and their interaction as fixed effects—GROUP (control vs. *Jyutping*), LIST (global vs. segmental) and GROUP: LIST. The raw RT was log-transformed to achieve closer approximation of normality of residuals. We also included standardized English scores ENGSCORE as a testing variable in all four models. For the random-effects structures of the four models, we started by including by-participant random intercept only. Then, we tested

whether the inclusion of by-participant random slope of LIST significantly improved model fit by the log-likelihood ratio test. According to the test results, the random slope was justified for all four models.

For the ERP data, we first calculated the global field power (GFP, i.e., spatial root mean square of voltage over all electrodes) on the grand-averaged waveforms for the phoneme discrimination task and pure tone discrimination task respectively according to the following formula:

$$GFP_t = \sqrt{\frac{\sum_{i=1}^{k} V_{it}^2}{k}}$$

Where t is time, k is the number of channels (k = 32), V_{it} is the voltage at channel i and time t.

For each task, a 50 ms time-window was centered at each peak of the GFP. For the phoneme discrimination task, the ERP data were analyzed in five time-windows—N1 (115~165 ms), P2 (215~265 ms), N2 (265~315 ms), P3 (395~445 ms) and Late Positive Component (LPC, 615~665 ms). As the GFP for the pure tone discrimination task revealed three auditory N1s elicited sequentially by each of the three tones in the tone-triplet, the three time-windows where auditory N1 response occurred were selected for analysis—N1(1) (85~135 ms), N1(2) (375~425 ms) and N1(3) (475~525 ms). Then, we selected three electrode groups (ELECGROUP) for statistical analysis: anterior (F1, F3, Fz, F4, F8), middle (T7, C3, Cz, C4, T8), and posterior (P7, P3, Pz, P4, P8). The grouping of electrodes gave sufficient spatial coverage to examine effects regarding scalp-recorded ERP components of interest. For each task, mean ERP amplitudes of each time-window were calculated for each participant, ELECGROUP and LIST. For each time-window (N1, P2, N2, P3, LPC in the phoneme task and N1(1), N1(2), N1(3) in the pure tone task), the ERP amplitudes were submitted to a three-way analysis of variance (ANOVA), with a between-subjects factor GROUP and two within-subjects factors LIST and ELECGROUP. The degrees of freedom were adjusted for deviance from sphericity using the Greenhouse–Geisser (GG) method when appropriate.

3. Results

3.1. Behavioral results

The model output is given in Fig. 2. For the phoneme discrimination task, the best ACC model revealed a main effect of GROUP (β =0.82, z = 2.8, $p < 0.01$), indicating that the *Jyutping* group achieved higher accuracy than the control group. There was also a significant main effect of LIST (β = -2.06, z = -8.0, $p < 0.001$), reflecting less accurate responses in the segmental list than in the global list. The best RT model revealed a significant main effect of LIST (β = 0.06, t = 4.6, $p < 0.001$), reflecting slower responses to the segmental list than to the global list. The absence of a significant interaction of GROUP by LIST in both models suggested that the *Jyutping* group performed better in phoneme discrimination regardless of list. For the pure tone discrimination task, no effects emerged as significant in the RT and ACC models. The testing variable ENGSCORE emerged as significant only in the RT model for the pure tone discrimination task, but it did not affect the significance of any other effects that we observed.

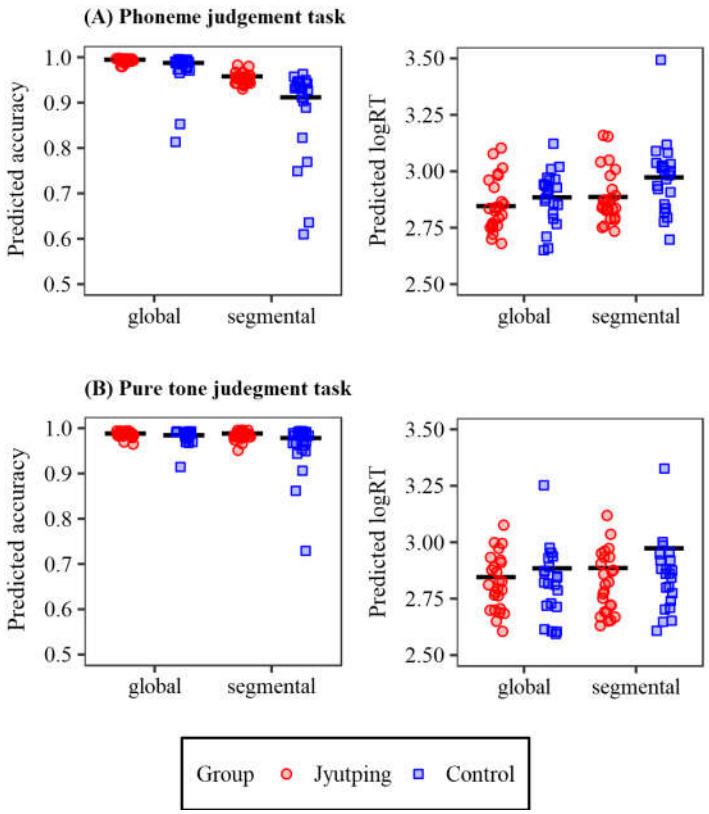

Fig. 2. Model predictions of ACC and logRT for (A) the phoneme discrimination task and (B) pure tone discrimination task. The solid line denotes estimated means based on fixed-effects while each dot represents individual fitted values based on fixed plus random effects.

3.2. ERP results

Fig. 3 displays the grand-averaged ERP waveforms at Cz and the topographic maps of N1, P2, N2, P3 and LPC for global and segmental lists respectively in the phoneme discrimination task.

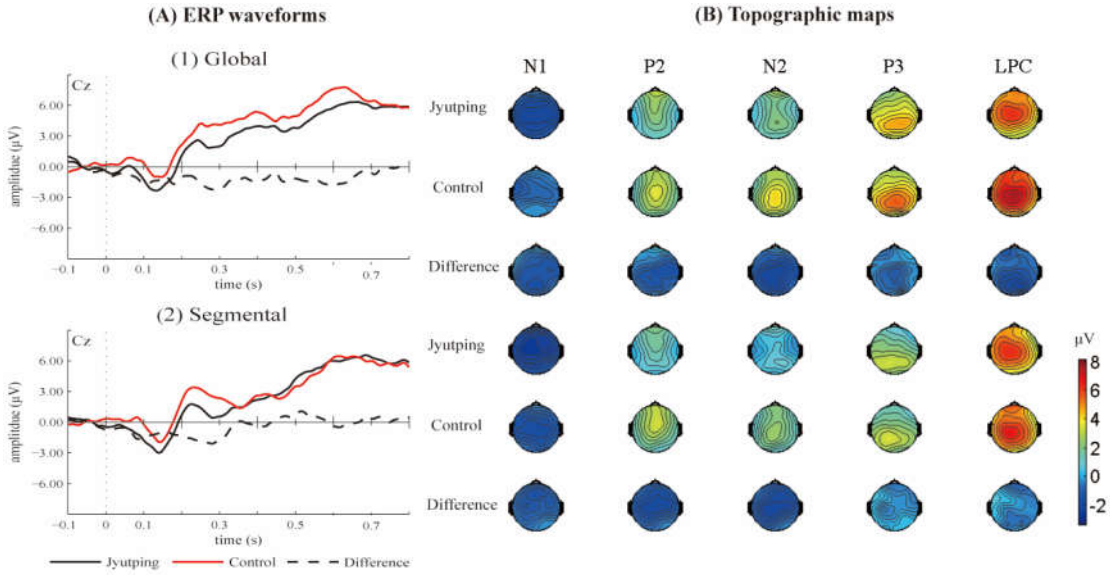

Fig. 3. Grand-averaged ERP waveforms (A) and topographic maps (B) for the phoneme discrimination task. The topographic maps are plotted based on the averaged amplitude across time points in five time-windows N1 (115~165 ms), P2 (215~265 ms), N2 (265~315 ms), P3 (395~445 ms) and LPC (615~665 ms) respectively.

The N1 amplitude for the phoneme discrimination task is presented in detail in Fig. 4. There was a significant main effect of GROUP ($F_{(1, 45)} = 5.71$, $p < 0.05$). Overall, the *Jyutping* group exhibited more negative N1 amplitude than the control group. There was also a main effect of LIST ($F_{(1, 45)} = 6.16$, $p < 0.05$), indicating more negative N1 amplitude elicited for the segmental list. We also found a significant main effect of ELECGROUP ($F_{(1.15, 51.88)} = 5.60$, $p < 0.05$, ; anterior electrodes = -1.38 μV, middle electrodes = -1.64 μV, posterior electrodes = -1.19 μV This result suggested that overall N1 has a fronto-central distribution. Most importantly, the three-way interaction among GROUP, ELECGROUP and LIST reached significance ($F_{(1.15, 52.11)} = 4.13$, $p < 0.05$). Post-hoc analysis revealed that for the global list the *Jyutping* group showed significantly larger N1 amplitude than the control group over the middle ($t_{(91.16)} = -2.27$, $p < 0.05$) and posterior electrode groups ($t_{(91.16)} = -2.26$, $p < 0.05$). For the segmental list the *Jyutping* group exhibited significantly larger N1 amplitude than the control group over the anterior ($t_{(91.16)} = -2.18$, $p < 0.05$) and middle electrode groups ($t_{(91.16)} = -2.11$, $p < 0.05$).

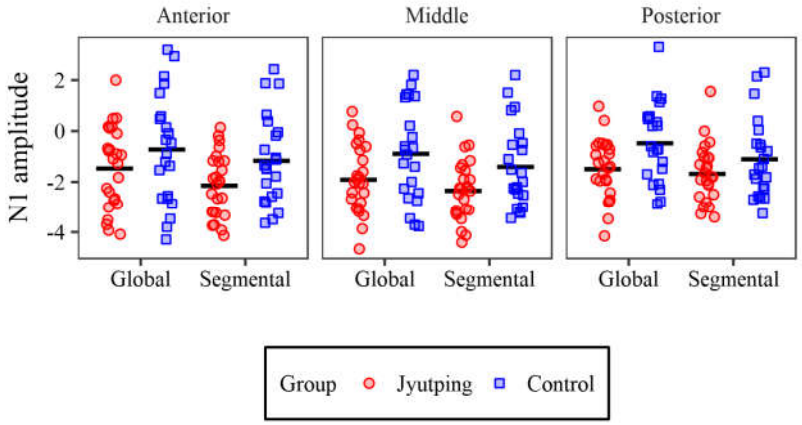

Fig. 4. N1 amplitude obtained in the phoneme discrimination task. The solid line denotes least square means estimated from the linear model while each dot represents individual N1 amplitude.

No GROUP effects were observed in later time-windows. For the P2 amplitude, no main effects or interactions emerged as significant. For the N2 amplitude, we found a main effect of LIST ($F_{(1, 45)} = 11.33$, $p < 0.01$), indicating more negative N2 amplitude elicited for the segmental list. For the P3 amplitude, there was a main effect of ELECGROUP ($F_{(1.2, 55.00)} = 22.88$, $p < 0.001$; anterior electrodes = 2.18 μV, middle electrodes = 3.04 μV, posterior electrodes = 3.75 μV). This was consistent with the centro-parietal topographic distribution for P3. There was also a main effect of LIST ($F_{(1,45)} = 21.08$, $p < 0.001$), with more negative amplitude elicited by the segmental list than by the global list. For LPC, a two-way interaction between LIST and ELECGROUP was found ($F_{(1.22, 54.91)} = 5.52$, $p < 0.05$). Post-hoc analysis did not reveal any list differences in the any electrode group, but the segmental list tended to elicit more negative LPC amplitude than the global list over anterior and middle electrode groups.

Fig. 5 shows the grand-averaged ERP waveforms at Cz and topographic maps of N1(1), N1(2) and N1(3) for the global and segmental lists respectively in the pure tone discrimination task.

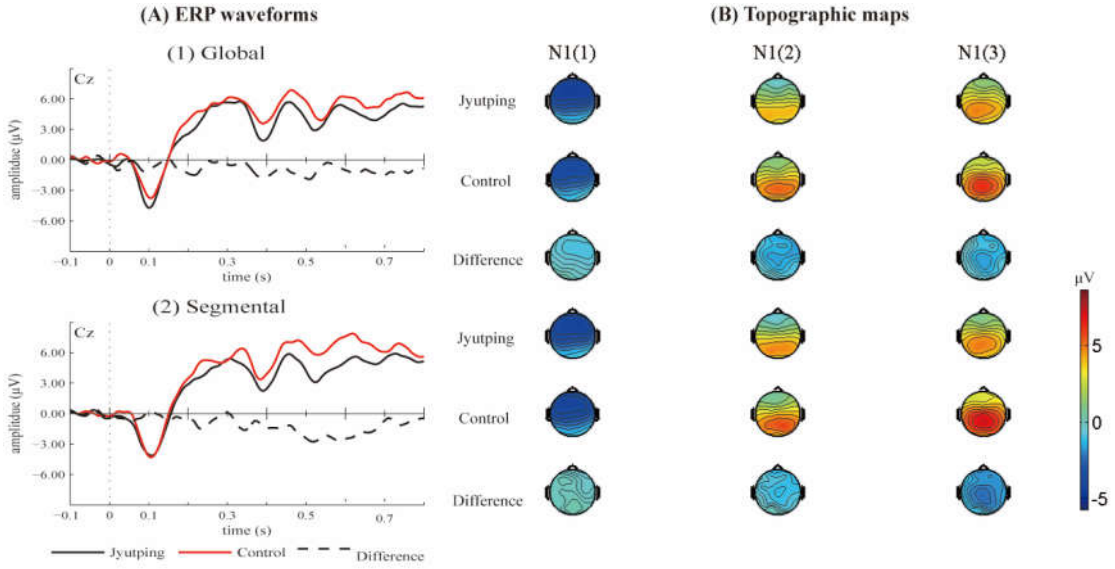

Fig. 5. Grand-averaged ERP waveforms (A) and topographic maps (B) for the pure tone discrimination task. The topographic maps are plotted based on the averaged amplitude across time points in three time-windows N1(1) (85~135 ms), N1(2) (375~425 ms) and N1(3) (475~525 ms) respectively.

The N1 amplitude for the pure tone discrimination task are presented in detail in Fig. 6. For the N1(1) amplitude, there was a main effect of ELECGROUP ($F_{(1.20, 54.02)}$ =83.76, $p < 0.001$; anterior electrodes = -3.91 μV, middle electrodes = -3.32 μV, posterior electrodes = -1.74 μV). This suggested that the N1(1) elicited in the pure tone task had a fronto-central scalp distribution.

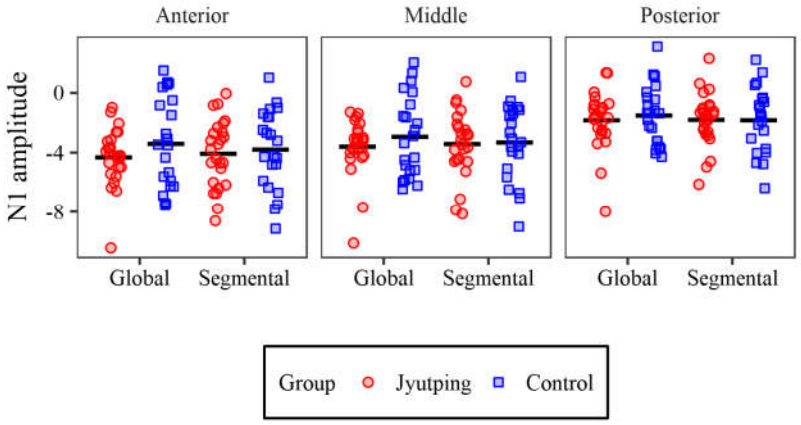

Fig. 6. N1(1) amplitude obtained in the pure tone discrimination task. The solid line denotes least square means estimated from the linear model while each dot represents individual N1 amplitude.

For the N1(2) amplitude, there was a significant main effect of ELECGROUP ($F_{(1.15, 52.08)}$ =66.43, $p < 0.001$; anterior electrodes = 1.28 μV, middle electrodes = 3.11 μV, posterior electrodes = 4.51 μV). This was broadly consistent with the fronto-central scalp distribution

of N1. ELECGROUP also interacted with LIST ($F_{(1.19, 53.70)} = 7.19$, $p < 0.01$). However, post-hoc analysis indicated no LIST difference at each level of ELECGROUP.

For the N1(3) amplitude, there was a significant main effect of ELECGROUP ($F_{(1.19, 53.59)} = 35.00$, $p < 0.0011$; anterior electrodes = 2.56 μV, middle electrodes = 4.12 μV, posterior electrodes = 4.84 μV). However, it is worth pointing out that both N1(2) and the later N1(3) appeared to be superimposed on a positive component with a centro-parietal scalp-distribution (see Fig. 5), which might reflect later perceptual or decisional processes. This possible component overlap would inevitably have some influence on the results in these two time-windows.

4. Discussion

4.1. Enhanced phonological awareness for logographic readers with alphabetic literacy skills

One purpose of the current study was to examine the impact of learning *Jyutping,* a native alphabetic script, on phonemic awareness in Cantonese-speaking logographic readers. We found that Cantonese-speaking logographic readers with proficiency in *Jyutping* performed more accurately than matched controls in the phoneme discrimination task but not in the pure tone discrimination task.

This finding was in line with the psycholinguistic grain size theory, which predicts that sensitivity to different levels of phonological units can be driven by the psycholinguistic grain sizes employed in phonology-orthography mapping (Ziegler & Goswami, 2005). Previous behavioral studies have shown that phonemic awareness skills are enhanced for logographic readers with alphabetic knowledge (de Gelder et al., 1993; Read et al., 1986) and that phonemic awareness primarily arises as a result of learning to read and write an alphabetic script (Morais et al., 1979). Our observations confirmed these previous findings and extended them by showing that psycholinguistic grain sizes are malleable through exposure to a novel orthography even in adulthood. Unlike the new generation of students in Mainland China who had received *Pinyin* instruction early in childhood, most of our *Jyutping* participants could not perform perfect sound-to-letter transcription. Nevertheless, their ability to analyze speech at a fine-grained phonemic level in their native language is boosted by acquiring a native alphabetic script in adulthood.

The absence of the group difference in the pure tone task confirmed that the two groups of participants were roughly equal in their ability to process non-speech materials in both global and segmental lists, suggesting that the alternations in psycholinguistic grain sizes brought about by enhanced alphabetic literacy skills might be restricted to the speech domain.

However, some caveats should also be noted. The group difference in response accuracy for the phoneme discrimination task seems to be small compared with previous studies (de Gelder et al., 1993; Read et al., 1986). In the phoneme discrimination task, both groups exhibited relatively high response accuracy and no group difference in reaction time was found. There are two possible explanations for this relatively small *Jyutping* advantage. First, this might be due to the strong influence of English orthography in Hong Kong. Note that the two groups were matched in their overall English ability. Although English scores did not predict their performance in the phoneme discrimination task, the group differences in judging phonemes could be diminished if both groups have developed some general phonological awareness skills through exposure to the alphabetic English orthography. Second, the strong group difference reported in previous studies could be due to confounding factors such as the age of participants. For instance, in Read et al. (1986), the *Pinyin* group was on average 16 years younger than the non-alphabetic group, which could contribute to a relatively large group difference. Nevertheless, as the tasks differed substantially across studies, further investigations using the same paradigm are needed to systematically assess

the degree to which the acquisition of *Jyutping*, *Pinyin*, or other alphabetic orthographies, affects phonological awareness in logographic readers.

The LIST effect also requires some explanation. Both groups achieved lower accuracy and responded more slowly in the segmental list than in the global list. This indicated that the segmental condition was more difficult for the participants, possibly due to increased segmentation effort and/or attentional processes in this list. Inconsistent with our prediction, our behavioral results revealed no interaction between GROUP and LIST. We will interpret this result below in light of the ERP data.

4.2. Group differences in N1 amplitude: attention-mediated phonetic/phonological processing?

ERP analysis allowed us to pinpoint the processing stage of the phoneme discrimination task that was influenced by alphabetic literacy skills. We found that the difference between the *Jyutping* and control groups emerged relatively early in time, that is, in the N1 time window. Overall, the segmental list elicited larger N1 amplitude than the global list, and, most critically, the *Jyutping* group showed larger N1 amplitude than the control group. This GROUP effect was mediated by LIST and ELECGROUP. In the global list, the GROUP effect was found in middle and posterior electrode groups whereas it emerged as significant in middle and anterior electrode groups for the segmental list. No significant GROUP or LIST effects were found for the ERP responses in the pure tone discrimination task.

Enhanced phonetic/phonological processing induced by the acquisition of a native alphabetic script is one possible account for the augmentation of N1 in the *Jyutping* group. Neuroimaging evidence suggests that brain regions responsible for phonetic/phonological processing, including the superior temporal gyrus (STG) and planum temporale (PT), show augmented activations in competent alphabetic orthography readers (Brennan et al., 2013; Dehaene et al., 2010, 2015). For ERPs, previous literature suggests that auditory N1 is associated with phonetic/phonological processing (Bonte & Blomert, 2004; Obleser et al., 2003), and thus group differences in phonetic/phonological processing might explain the N1 GROUP effect in our study. Furthermore, PT and STG have been reported to be probable neural generators for auditory N1 (Eggermont & Ponton, 2002; Wolpaw & Penry, 1975), further corroborating the phonological/phonetic processing account.

It is also likely to further attribute this group difference in N1 amplitude to differential attentional modulation related to phonetic/phonological processing in the two groups. Both meta-phonological tasks employed in Burton et al. (2000) and in the current study probably involved mechanisms for auditory temporal selective attention to phonological units. Participants were asked to judge whether the initial phonemes of two pseudowords were identical. Thus, they were required to voluntarily focus their temporal selective attention on two syllable-initial moments where the acoustic cues for the onset occurred.

More evidence for the attentional modulation account comes from previous ERP studies on the effect of selective attention on early sensory processing. It has been shown that spatial attention affects early sensory processing. Targets presented in attended spatial locations, i.e., left or right visual field for vision (Van and Hillyard, 1977) and left or right ear for audition (Hillyard et al., 1973; Woldorff & Hillyard, 1991) elicited larger early ERP responses, like N1, than those in unattended locations. Similar to spatial attentional effects, temporal selective attention to visual or auditory targets presented in a specific time interval, e.g., early or late interval, has also been found to enhance early visual (Correa et al., 2005) and auditory processing (Lange et al., 2003). All these findings can be taken as evidence for an early attentional modulation of sensory processing. Furthermore, sustained selective attention can evoke a processing negativity (PN) or negative difference (Nd), which typically develops prior to stimulus onset and extends into stages of early sensory processing (Lange et al., 2003; Näätänen et al., 1978). In light of these findings, the increased N1 amplitude observed in the

logographic readers with alphabetic script expertise during initial phoneme discrimination could also reflect an attention-induced enhancement of phonetic/phonological processing or a superimposition of an attention-related PN or Nd before and/or during the attended moment in time. It should be noted that although we only found a significant GROUP effect on the N1 amplitude, the difference waveforms and topographies (see Fig. 3) indeed exhibit a sustained negativity for the *Jyutping* group beyond the N1 time-window.

There is also some neuroimaging evidence demonstrating the engagement of attentional networks in meta-phonological tasks, such as pseudoword repetition (Castro-Caldas et al., 1998; Petersson et al., 2000) and phoneme discrimination (Burton et al., 2000). Castro-Caldas et al., (1998) and Petersson et al. (2000) further reported that the literate and illiterate differed in functional activations of the attentional network in pseudoword repetition.

The phonetic/phonological processing and attention modulation accounts focus on different facets of the N1 enhancement, but they are probably not mutually exclusive and may even be deemed complementary to each other. According to the phonetic/phonological account, the N1 enhancement is brought about by mental operations specific to phonetic/phonological processing, like phonetic feature extraction and phonological category access, which are confined to the speech domain and not involved in domain-general auditory processing. This might be one reason why group differences were not observed in the pure tone discrimination task. However, the phonetic/phonological account does not specify whether and how attentional mechanisms might mediate these phonetic/phonological processes. It is possible that attention itself or its interaction with other phonetic/phonological processes can explain the N1 enhancement. As discussed above, the N1 enhancement observed in logographic readers with higher alphabetic literacy skills might be interpreted as sharpened temporal selective attention to phonological units that unfold in time. This interpretation fits nicely with the concept of ‘phonological awareness’, which emphasizes the domain-specific awareness/attention to phonological units. It is also consistent with the absence of group effects in the pure tone discrimination task.

The overall larger N1 amplitude elicited in the segmental list than in the global list can be interpreted as the increased attentional mediation under this condition. This explanation is broadly consistent with Burton et al.'s (2000) finding that the anterior cingulate cortex exhibited stronger activations for the segmental list than for the global list in the phoneme discrimination task, presumably reflecting a higher requirement of attention or conflict monitoring for the former list.

In addition to these main effects of GROUP and LIST mentioned above, we also found that the electrodes showing N1 amplitude enhancement in the *Jyutping* group varied across the two lists. The enhancement was present at middle and posterior sites for the global list whereas it was found at anterior and middle electrode sites for the segmental list. The fronto-central GROUP effect observed in the segmental list can be interpreted as being caused by the modest recruitment of frontal neural sources for the control group. In meta-phonological tasks especially for the segmental list, additional frontal sources are hypothesized to be associated with phonemic segmentation, articulatory recoding, working memory and/or attentional processes (Burton et al., 2000; Castro-Caldas et al., 1998; Petersson et al., 2000; Zatorre et al., 1996). The *Jyutping* group might engage these frontal neural substrates more efficiently than the control group due to their heightened phonetic/phonological processing and attentional skills induced by the acquisition of a native alphabetic script. The centro-parietal GROUP effect observed in the global list is more difficult to interpret. It seems that for the global list the *Jyutping* participants also engaged some additional neuro-cognitive processes, which are presumably related to phoneme transcription and segmentation. Although it is assumed that no phoneme transcription and segmentation are explicitly needed for the global list, we cannot prevent *Jyutping* users from actively engaging some of these

processes under this condition. Nevertheless, the exact functional significance of these additional neural activities remains murky. They might pertain to other sub-processes in the attentional network. Another speculative explanation is that the enhancement of centro-parietal activity might result from the group differences in the co-activation of orthographic representations subserved by the occipito-temporal region (VWFA), as these orthographic processes have been consistently reported to play a critical role in meta-phonological tasks (Burton et al., 2000; Lafontaine et al., 2012). Yet, whether group differences in orthographic activations can occur as early as in the N1 time-window or whether other neuro-cognitive processes contribute to group differences in scalp-recorded N1 for the global list needs to be addressed by future studies. Given the N1 GROUP effects for both lists, one last point regarding the lack of GROUP:LIST interaction in the behavioral data needs to be made. As mentioned earlier, the behavioral results revealed no interaction between GROUP and LIST, which deviates from the prediction based on the results of Burton et al., (2000). It is possible that behavioral measures are not fine-grained and sensitive enough to reveal this interaction. Indeed, the ERP analyses revealed different spatial features over the scalp for different lists for the N1 group effect. The potentially different neural resources engaged in the segmental and global lists likely contributed to similar group effects behaviorally.

Our findings also have implications for the nature of phonological awareness and its relations to other cognitive capacities. Most theories concerning phonological awareness, e.g., the psycholinguistic grain size theory, are built upon behavioral evidence and basically conceptualize phonological awareness as a somewhat unitary notion in phonological and reading development. We suggest that phonological awareness is probably not a unitary concept or at least it is closely associated with multiple cognitive mechanisms. Our current observations and previous neuroimaging results have pointed to the role of attention in phonological awareness (Burton et al., 2000; Petersson et al., 2000). While the relationship between attention and phonological awareness remains controversial, these two concepts are intricately related to each other (De Brigard & Prinz, 2010). More specifically for phonological awareness as measured by certain meta-phonological tasks, capturing smaller phonological units that unfold in time may require auditory temporal attention specifically to established phonological units. In addition, as the awareness of smaller phonological units, like phonemes, per se is largely shaped by visual experiences with alphabetic orthography (Dehaene et al., 2015; Morais et al., 1979), orthographic representations and processes are crucial underpinnings of phonological awareness. Establishing the link between phonological awareness and other cognitive abilities is important as it may further our understanding of the origins of some phonological deficits, e.g., those manifested in development dyslexia (Bonte & Blomert, 2004; Ziegler & Goswami, 2005). We await future studies to provide a clearer picture of the complicated interactions between these components relating to phonological awareness.

4.3. N2, P3 and LPC: higher-level cognitive processing

In the phoneme discrimination task, no GROUP effects reached significance in later N2, P3 and LPC time-windows, which are assumed to reflect post-perceptual higher-level cognitive processes. However, the significant LIST effects found in these time-windows, i.e., larger N2 amplitude and smaller P3 amplitude elicited for the segmental list than for the global list, need some explanations. In addition, there was a similar trend that LPC was smaller in the segmental list.

N2 is hypothesized to index automatic mismatch detection and cognitive control, like response inhibition (Folstein & Van Petten, 2008). The N2 LIST effect can be explained by either mismatch detection or inhibitory control. First, as the rimes differed in the segmental list as a potential distractor, the perceptual mismatch between the first and the second

pseudowords also increased, presumably leading to the enlarged N2 amplitude for the segmental list. This component can probably be linked to the phonological mismatch negativity (PMN), which may be present whenever a phonological mismatch between expectations and acoustic traces occurs (Bonte & Blomert, 2004; Lafontaine et al., 2012; Zhang, 2017). Second, response inhibition might be another potential explanation for this effect. The segmental list required a higher level of inhibition process, as the participants had to inhibit the salient rime interference to achieve a better discrimination of the initial phonemes.

The P3 family is thought to consist of subcomponents such as P3a and P3b (Picton, 1992; Polich, 2007). P3a and P3b are assumed to be related to cognitive processes such as attention allocation, memory updating, or stimulus categorization (Azizian et al., 2006; Polich, 2007). The relatively small P3 amplitude elicited in the segmental list presumably reflects the increased difficulty of stimulus categorization, i.e., the evaluation of the task-relevant initial consonants with rime interference.

For LPC, its functional role in human information processing is disputed (Bakhtiar et al., 2016; Friedman, 1984). Some studies suggest that LPC indexes some level of decisional processes (Bornkessel-Schlesewsky et al., 2011; Finnigan et al., 2002). Following this interpretation, the trend for a relatively small LPC elicited in the segmental list might be interpreted as more difficult decisional processes or reduced decisional confidence under this condition.

5. Conclusion

In this ERP study, we examined how alphabetic literacy skills influence the time course of initial phoneme discrimination by comparing logographic Chinese readers with high and little *Jyutping* expertise. We found that Chinese readers with *Jyutping* expertise performed the task more accurately than their peers. This result supports the existing claim that alphabetic literacy skills boost phonemic awareness in non-alphabetic readers. Most importantly, the ERP results indicate that in this meta-phonological task, literacy-related difference in ERP amplitude was detected as early as in the N1 time window. We suggest that the enhanced auditory N1 might reflect the increased efficiency of phonetic-phonological processing in logographic readers with high alphabetic skills. This improvement may be mediated by attentional processes, i.e., logographic readers with alphabetic knowledge of their native language are more capable of selectively attending to phonological segments that unfold in time. We argue that phonological awareness is a multi-faceted linguistic ability, of which selective attention is likely to be an important aspect. In addition, the changes in neuro-cognitive processes following the acquisition of a native alphabetic script can occur relatively late in adulthood for logographic readers, which underscores the plasticity of the human brain even in adulthood.

Compliance with ethical standards

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Conflict of Interest: Yubin Zhang declares that he has no conflict of interest. Chotiga Pattamadilok declares that she has no conflict of interest. Dustin Kai Yan Lau declares that he has no conflict of interest. Mehdi Bakhtiar declares that he has no conflict of interest. Long-Ying Yim declares that she has no conflict of interest. Ka-Yui Leung declares that she has no conflict of interest. Caicai Zhang declares that she has no conflict of interest.

Ethical approval: All procedures performed in studies involving human participants were in accordance with the ethical standards of the Human Subjects Ethics Sub-committee of the Hong Kong Polytechnic University and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent: Informed consent was obtained from all individual participants included in the study.

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Fig. 7. The schematic flow diagram of one experimental trial. The underlined portions denote the initial consonants and the 1st pure tones, to which participants were required to make 'same' or 'different' responses.

Fig. 8. Model predictions of ACC and logRT for (A) the phoneme discrimination task and (B) pure tone discrimination task. The solid line denotes estimated means based on fixed-effects while each dot represents individual fitted values based on fixed plus random effects.

Fig. 9. Grand-averaged ERP waveforms (A) and topographic maps (B) for the phoneme discrimination task. The topographic maps are plotted based on the averaged amplitude across time points in five time-windows N1 (115~165 ms), P2 (215~265 ms), N2 (265~315 ms), P3 (395~445 ms) and LPC (615~665 ms) respectively.

Fig. 10. N1 amplitude obtained in the phoneme discrimination task. The solid line denotes least square means estimated from the linear model while each dot represents individual N1 amplitude.

Fig. 11. Grand-averaged ERP waveforms (A) and topographic maps (B) for the pure tone discrimination task. The topographic maps are plotted based on the averaged amplitude across time points in three time-windows N1(1) (85~135 ms), N1(2) (375~425 ms) and N1(3) (475~525 ms) respectively.

Fig. 12. N1(1) amplitude obtained in the pure tone discrimination task. The solid line denotes least square means estimated from the linear model while each dot represents individual N1 amplitude.

Supplemental materials. Experimental materials

Table S1. Six sets of Cantonese pseudowords in the phoneme discrimination task.

set 1	aa	aai	set 2	ei	eoi
p	paa2	paai2	d	dei4	deoi4
f	faa2	faai2	g	gei4	geoi4
set 3	aau	i	set 4	it	aat
m	maau2	mi2	c	cit1	caat1
l	laau2	li2	l	lit1	laat1
set 5	in	eng	set 6	ek	ok
g	gin4	geng4	t	tek1	tok1
h	hin4	heng4	s	sek1	sok1

Table S2. Six sets of tone-triplets in the pure tone discrimination task.

set 1	$1600_{Hz}800_{Hz}$	$1400_{Hz}1800_{Hz}$	set 2	$200_{Hz}1200_{Hz}$	$600_{Hz}2000_{Hz}$
200_{Hz}	200_{Hz}-$1600_{Hz}800_{Hz}$	200_{Hz}-$1400_{Hz}1800_{Hz}$	1000_{Hz}	1000_{Hz}-$200_{Hz}1200_{Hz}$	1000_{Hz}-$600_{Hz}2000_{Hz}$
1000_{Hz}	1000_{Hz}-$1600_{Hz}800_{Hz}$	1000_{Hz}-$1400_{Hz}1800_{Hz}$	2200_{Hz}	2200_{Hz}-$200_{Hz}1200_{Hz}$	2200_{Hz}-$600_{Hz}2000_{Hz}$
set 3	$800_{Hz}2200_{Hz}$	$600_{Hz}1400_{Hz}$	set 4	$2000_{Hz}600_{Hz}$	$200_{Hz}1400_{Hz}$
1600_{Hz}	1600_{Hz}-$800_{Hz}2200_{Hz}$	1600_{Hz}-$600_{Hz}1400_{Hz}$	400_{Hz}	400_{Hz}-$2000_{Hz}600_{Hz}$	400_{Hz}-$200_{Hz}1400_{Hz}$
200_{Hz}	200_{Hz}-$800_{Hz}2200_{Hz}$	200_{Hz}-$600_{Hz}1400_{Hz}$	1800_{Hz}	1800_{Hz}-$2000_{Hz}600_{Hz}$	1800_{Hz}-$200_{Hz}1400_{Hz}$
set 5	$1000_{Hz}200_{Hz}$	$600_{Hz}2200_{Hz}$	set 6	$1400_{Hz}600_{Hz}$	$200_{Hz}1200_{Hz}$

800_{Hz}	800_{Hz}-1000_{Hz}	800_{Hz}-600_{Hz} 2200_{Hz}	1800_{Hz}	1800_{Hz}-1400_{Hz}	1800_{Hz}-200_{Hz} 1200_{Hz}
2000_{Hz}	2000_{Hz}-1000_{Hz}	2000_{Hz}-600_{Hz} 2200_{Hz}	400_{Hz}	400_{Hz}-1400_{Hz}	400_{Hz}-200_{Hz} 1200_{Hz}